# DBMS LAB ASSIGNMENT-2

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## Database setup

```
CREATE DATABASE DBMS_LAB_ASSIGNMENT2;
USE DBMS_LAB_ASSIGNMENT2;
```

## QUESTION 1: CREATE EMP1 TABLE

**MYSQL COMMAND:**

```
CREATE TABLE EMP1 (
ID INT PRIMARY KEY,
Name VARCHAR(10),
Basic DECIMAL(6,2),
Designation VARCHAR(10),
Age INT
);

DESC EMP1;
```

**EXPECTED MYSQL OUTPUT:**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| ID | int | NO | PRI | NULL | |
| Name | varchar(10) | YES | | NULL | |
| Basic | decimal(6,2) | YES | | NULL | |
| Designation | varchar(10) | YES | | NULL | |
| Age | int | YES | | NULL | |

## QUESTION 2: CHANGE DATA TYPE OF BASIC

**MYSQL COMMAND:**

```
ALTER TABLE EMP1
MODIFY Basic INT;

DESC EMP1;
```

**EXPECTED MYSQL OUTPUT:**

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| ID | int | NO | PRI | NULL | |
| Name | varchar(10) | YES | | NULL | |
| Basic | int | YES | | NULL | |
| Designation | varchar(10) | YES | | NULL | |
| Age | int | YES | | NULL | |

## QUESTION 3: CHANGE NAME SIZE

### MYSQL COMMAND:

```
ALTER TABLE EMP1
MODIFY Name VARCHAR(15);

DESC EMP1;
```

### EXPECTED MYSQL OUTPUT:

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| ID | int | NO | PRI | NULL | |
| Name | varchar(15) | YES | | NULL | |
| Basic | int | YES | | NULL | |
| Designation | varchar(10) | YES | | NULL | |
| Age | int | YES | | NULL | |

## QUESTION 4: CREATE EMP_trainee TABLE

### MYSQL COMMAND:

```
CREATE TABLE EMP_trainee (
Emp_id INT,
Name VARCHAR(15),
Basic INT,
Designation VARCHAR(10),
Age INT
);

DESC EMP_trainee;
```

### EXPECTED MYSQL OUTPUT:

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| Emp_id | int | YES | | NULL | |
| Name | varchar(15) | YES | | NULL | |
| Basic | int | YES | | NULL | |
| Designation | varchar(10) | YES | | NULL | |
| Age | int | YES | | NULL | |

## QUESTION 5: INSERT DATA INTO EMP1

**MYSQL COMMAND:**

```
INSERT INTO EMP1 (ID, Name, Basic, Designation, Age)
VALUES
(1, 'Rohit', 6700, 'Manager', 24),
(2, 'Sunil', 6200, 'Engineer', 27),
(3, 'Payal', 6300, 'Engineer', 25),
(4, 'Kunal', 6700, 'Trainee', 28),
(5, 'Sunita', 6230, 'Trainee', 26),
(6, 'Bimal', 7000, 'Trainee', 25);

SELECT * FROM EMP1;
```

**EXPECTED MYSQL OUTPUT:**

| ID | Name | Basic | Designation | Age |
|---|---|---|---|---|
| 1 | Rohit | 6700 | Manager | 24 |
| 2 | Sunil | 6200 | Engineer | 27 |
| 3 | Payal | 6300 | Engineer | 25 |
| 4 | Kunal | 6700 | Trainee | 28 |
| 5 | Sunita | 6230 | Trainee | 26 |
| 6 | Bimal | 7000 | Trainee | 25 |

## QUESTION 6: INSERT TRAINEE RECORDS

**MYSQL COMMAND:**

```
INSERT INTO EMP_trainee (Emp_id, Name, Basic, Designation, Age)
SELECT ID, Name, Basic, Designation, Age
FROM EMP1
WHERE Designation = 'Trainee';

SELECT * FROM EMP_trainee;
```

**EXPECTED MYSQL OUTPUT:**

| Emp_id | Name | Basic | Designation | Age |
|---|---|---|---|---|
| 4 | Kunal | 6700 | Trainee | 28 |
| 5 | Sunita | 6230 | Trainee | 26 |
| 6 | Bimal | 7000 | Trainee | 25 |

## QUESTION 7: ADD SKILLS AND DOJ COLUMNS

### MYSQL COMMAND:

```
ALTER TABLE EMP1
ADD Skills VARCHAR(10),
ADD DOJ DATE;

DESC EMP1;
```

### EXPECTED MYSQL OUTPUT:

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| ID | int | NO | PRI | NULL | |
| Name | varchar(15) | YES | | NULL | |
| Basic | int | YES | | NULL | |
| Designation | varchar(10) | YES | | NULL | |
| Age | int | YES | | NULL | |
| Skills | varchar(10) | YES | | NULL | |
| DOJ | date | YES | | NULL | |

## QUESTION 7 (CONTINUED): UPDATE SKILLS AND DOJ VALUES

### MYSQL COMMAND:

```
UPDATE EMP1 SET Skills = 'Manage', DOJ = '2020-01-15' WHERE ID = 1;
UPDATE EMP1 SET Skills = 'Develop', DOJ = '2020-02-20' WHERE ID = 2;
UPDATE EMP1 SET Skills = 'Testing', DOJ = '2020-03-10' WHERE ID = 3;
UPDATE EMP1 SET Skills = 'Manage', DOJ = '2020-04-15' WHERE ID = 4;
UPDATE EMP1 SET Skills = 'Design', DOJ = '2020-05-20' WHERE ID = 5;
UPDATE EMP1 SET Skills = 'Develop', DOJ = '2020-06-25' WHERE ID = 6;

SELECT * FROM EMP1;
```

### EXPECTED MYSQL OUTPUT:

| ID | Name | Basic | Designation | Age | Skills | DOJ |
|---|---|---|---|---|---|---|
| 1 | Rohit | 6700 | Manager | 24 | Manage | 2020-01-15 |
| 2 | Sunil | 6200 | Engineer | 27 | Develop | 2020-02-20 |
| 3 | Payal | 6300 | Engineer | 25 | Testing | 2020-03-10 |
| 4 | Kunal | 6700 | Trainee | 28 | Manage | 2020-04-15 |
| 5 | Sunita | 6230 | Trainee | 26 | Design | 2020-05-20 |
| 6 | Bimal | 7000 | Trainee | 25 | Develop | 2020-06-25 |

## QUESTION 8: UPDATE MORE THAN ONE ROW IN ONE QUERY

### MYSQL COMMAND:

```
SET SQL_SAFE_UPDATES = 0;

UPDATE EMP1
SET Basic = Basic + 500
WHERE Designation = 'Trainee';

SET SQL_SAFE_UPDATES = 1;

SELECT * FROM EMP1;
```

### EXPECTED MYSQL OUTPUT:

| ID | Name | Basic | Designation | Age | Skills | DOJ |
|---|---|---|---|---|---|---|
| 1 | Rohit | 6700 | Manager | 24 | Manage | 2020-01-15 |
| 2 | Sunil | 6200 | Engineer | 27 | Develop | 2020-02-20 |
| 3 | Payal | 6300 | Engineer | 25 | Testing | 2020-03-10 |
| 4 | Kunal | 7200 | Trainee | 28 | Manage | 2020-04-15 |
| 5 | Sunita | 6730 | Trainee | 26 | Design | 2020-05-20 |
| 6 | Bimal | 7500 | Trainee | 25 | Develop | 2020-06-25 |

## QUESTION 9: RENAME AGE COLUMN

### MYSQL COMMAND:

```
ALTER TABLE EMP1
RENAME COLUMN Age TO Age_in_Years;

DESC EMP1;
```

### EXPECTED MYSQL OUTPUT:

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| ID | int | NO | PRI | NULL | |
| Name | varchar(15) | YES | | NULL | |
| Basic | int | YES | | NULL | |
| Designation | varchar(10) | YES | | NULL | |
| Age_in_Years | int | YES | | NULL | |
| Skills | varchar(10) | YES | | NULL | |
| DOJ | date | YES | | NULL | |

## QUESTION 10: DROP AGE COLUMN FROM EMP_trainee

### MYSQL COMMAND:

```
ALTER TABLE EMP_trainee
DROP COLUMN Age;

DESC EMP_trainee;
```

### EXPECTED MYSQL OUTPUT:

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| Emp_id | int | YES | | NULL | |
| Name | varchar(15) | YES | | NULL | |
| Basic | int | YES | | NULL | |
| Designation | varchar(10) | YES | | NULL | |

## QUESTION 11: RENAME TABLE EMP1

### MYSQL COMMAND:

```
RENAME TABLE EMP1 TO EMP_Mgr_Engr;

SHOW TABLES;
```

### EXPECTED MYSQL OUTPUT:

| Tables_in_DBMS_LAB_ASSIGNMENT2 |
|---|
| EMP_Mgr_Engr |
| EMP_trainee |

## QUESTION 12: TRUNCATE EMP_Mgr_Engr TABLE

### MYSQL COMMAND:

```
TRUNCATE TABLE EMP_Mgr_Engr;

SELECT * FROM EMP_Mgr_Engr;
```

### EXPECTED MYSQL OUTPUT:

| Empty set (0.00 sec) |
|---|